relays.weebly.com

What Is a Control Relay?

5-6 minutes

Control relays are an extremely important electrical part, and they play a vital role in today's electronic devices. A control relay is, in essence, a

switch that is controlled by electrical current

. A control relay is an electrical component that opens or closes a switch in order to allow current to flow through a conducting coil, with the coil not coming into direct contact with the switch. Control relays are electromagnetic devices that typically control the power flow in circuits. Control relays were invented more than a century ago, and today's world of electronics is dependent on control relays to control the currents that power so many of modern society's electric and electronic gadgets. Control relays are electronic parts that are used in motors, power plants, power supply systems, transistors, and more.

Control relays permit a low current circuit to control a high current circuit. Control relays are coils that have electricity sent through them. When electrical current is sent through the coil, the current generates an electromagnetic field that then works to operate the electrical device. Control relays have two methods in which they

operate. First, when electrical current flows through a coil, it draws in a magnetic substance that closes the device's switch. When the current stops flowing through the coil, the switch opens. A spring helps in this operation of switches, and electronic circuits are coupled to control relays in order to make devices operate automatically, and the switch is pre-programmed when needed. There are a few different types of relays used in today's electric and electronic applications such as single- and dual-switch relays. As electrical current flows through a coil this behavior creates magnetic poles. These are known, aptly, as North Pole and South Pole. An armature is coupled to a spring inside, to which the poles are attracted. When the current in the coil is at a satisfactory level, the armature remains attracted to the coil. In this situation, the switch stays on, and when the electrical current decreases, the attraction makes the spring pull the armature back, bringing the switch to the off position. Control relays are affected by temperatures, and their life cycles are negatively affected if temperatures are too high.

In today's world of electronics there are many types of control relays, and categorization often depends on sizes and specifications. Some of the main types of relays are solid state, monitoring, safety, time delay, general purpose, and latching relays.

Unlike most relays,

solid state relays

do not contain moving parts. Solid state relays offer quicker

operation than electromechanical relays. However, they suffer from lower and higher resistance in open and closed positions. Solid state relays are used in many of today's electronic devices such as amplifiers and other audio equipment. Aptly named, monitoring relays are used for monitoring various conditions. Voltage, temperature, and electrical current are a few of the conditions that will cause safety issues. Monitoring relays are used to protect expensive electronic equipment as they provide protections because they monitor harmful situations. Safety relays, like monitoring relays, are designed to provide protection. They typically are used for safety circuits for use in industrial equipment. Time-delay relays, also called time-release relays, are designed to stay on for a required time period. Time-delay relays permit actions to occur in electrical devices because they act as a timer. These relays are used when it is necessary to stop or start electrical current from moving into coils and armatures, and they permit electrical circuits to release at specified times. Conveyor belts are a good example of electrical devices that use time-delay relays, as they need certain actions to happen at certain times, with one part of a conveyor belt starting after another part stops. General purpose relays are just what their name implies. They are a generic type of relay that is built to provide circuit control within an application that doesn't necessarily require specific characteristics such as specified control of flow. Latching relays, or impulse relays as they are also referred to, are electromechanical switches that control larger flows of current with smaller flows. These provide two relaxed states which allow the relay to stay in its last position once the current is switched to the off position. The relay remains so and will not alter its state until another electrical current is applied.